

Part 14

Upload Files in Django

VERSION: 4.0.1

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How to upload Files in Django.?

We all have used social platforms like Instagram and Facebook. The one thing common in all is that users can upload images, videos on them. We upload our files on a server and then others view it. That is the whole concept of social media.

So, how Django upload a file? We will be answering the same question in this article. The contents of this article are:

- 1) How to upload file in Django.
- 2) Configuring settings.
- 3) A User Profile Application.

Configuring Settings

Before any file upload application, these settings are necessary. Open project's **settings.py** file and add these lines below the **STATIC FILES** settings.

```
MEDIA_ROOT = os.path.join(BASE_DIR, 'media')  
MEDIA_URL = 'media/'
```

Make a Directory named Media

Now, make a directory named **media** at the project level.

Media directory exists at the same level as **manage.py** file.

Explanation

MEDIA_ROOT : is an empty string by default. It takes in the absolute file path of a directory. We have used the same join functions in *Django static files tutorial* that we have passed in the file path of media directory. This directory will store all the files a user uploads in Django.

MEDIA_URL works the same way as **STATIC_URL**. This URL is used by Django instead of serving files with an absolute path. When Django will serve an uploaded file media it will be automatically added to the URL.

Make Model For Image

Now, paste this code in `profile_maker/models.py` file.

```
from django.db import models

class User_Profile(models.Model):
    fname = models.CharField(max_length=200)
    lname = models.CharField(max_length = 200)
    technologies = models.CharField(max_length=500)
    email = models.EmailField(default = None)
    display_picture = models.FileField()

    def __str__(self):
        return self.fname
```

display_picture is a **File Field**. It should have been an image field. We use a file field to store files on the server. It will create a File object which has many built-in attributes and methods. When a user uploads a file in the file field, we are storing file object in database and store file in **MEDIA_ROOT**.

Make a Form

Make a new forms.py file in your application. Inside **profile_maker/forms.py** file, paste this code:

```
from django import forms
from .models import User_Profile

class Profile_Form(forms.ModelForm):

    class Meta:
        model = User_Profile
        fields = [ 'fname', 'lname', 'technologies', 'email', 'display_picture' ]
```

Make Views

Now, in `profile_maker/views.py` file, paste this.

```
from .forms import Profile_Form
from .models import User_Profile

IMAGE_FILE_TYPES = ['png', 'jpg', 'jpeg']

def create_profile(request):
    form = Profile_Form()
    if request.method == 'POST':
        form = Profile_Form(request.POST, request.FILES)
        if form.is_valid():
            user_pr = form.save(commit=False)
            user_pr.display_picture = request.FILES['display_picture']
            file_type = user_pr.display_picture.url.split('.')[-1]
            file_type = file_type.lower()
            if file_type not in IMAGE_FILE_TYPES:
                return render(request, 'profile_maker/error.html')
            user_pr.save()
            return render(request, 'profile_maker/details.html', {'user_pr': user_pr})
    context = {"form": form,}
    return render(request, 'profile_maker/create.html', context)
```

Create Template

Now, again make a templates folder. The directory pattern will be **profile_maker/templates/profile_maker**. Remember to make two directories. First is templates and then the profile_maker directory inside it.

```
<form method="POST" enctype="multipart/form-data">
  <!-- Very Important csrf Token -->
  {% csrf_token %}
  <table>
    {{ form.as_table }}
  </table>
  <input type="submit" class="btn btn-primary btn-lg" name="register">
</form>
```


Configure URL

We have made all the files. Open your main urls.py file. In this file, paste this code:

```
from myproject import settings
from django.conf.urls.static import static

urlpatterns = [
    path('', views.create_profile, name = 'create'),
]

if settings.DEBUG:
    urlpatterns += static(settings.STATIC_URL, document_root = settings.STATIC_ROOT)
    urlpatterns += static(settings.MEDIA_URL, document_root = settings.MEDIA_ROOT)
```

Thanks for Reading